

Final Draft: “Is healthcare more expensive for cisgender Men vs Women and how so?”

Introduction

Cisgender men and women interact with the healthcare system in distinct ways, shaped by behavioural patterns, biological differences and social factors. These differences can lead to unequal health needs, outcomes and NHS spending. Yet, public datasets rarely offer a clear, integrated view of how healthcare costs vary by biological sex across the life course.

Our project aims to create a clear data driven comparison of cisgender men vs women healthcare costs by combining multiple NHS and ONS datasets. Using Jupyter Notebook, we will build a narrative dashboard that highlights how costs evolve over time, which conditions drive them, and where utilisation patterns differ.

Why this problem matters (and why it’s realistic)

Public NHS and ONS datasets are easily accessible, structured and well-documented. This makes the project feasible within the timeline and well-suited to the skills taught in the Data Specialisation module, such as data cleaning, joining datasets, dashboarding, and storytelling.

Analysing cisgender-based cost differences helps to illuminate:

- Differences in healthcare utilisation (including use of private healthcare referrals. providers)
- Variations in risk factors (location, lifestyle, age)
- Potential inequities or unmet needs (help-seeking behavioural patterns in data)
- Areas where one sex faces a greater medical or financial burden.

Understanding these patterns supports better NHS resource planning and service design.

How our project tackles the problem

We aim to answer three core questions:

1. Do cisgender men or women incur higher overall NHS Healthcare costs?
2. Which conditions, services, or age bands contribute most to these differences?
3. How do utilisation patterns differ (e.g. preventative vs emergency care)

To answer these, we will:

- Compare cisgender men vs women costs across conditions, age groups and service categories

- Analyse usage patterns in data such as:
 - ❖ GP visits
 - ❖ A&E Activity
 - ❖ Maternity care
 - ❖ Trauma-related care
 - ❖ Long-term conditions
- Link cost drivers to demographic and lifestyle factors (e.g. lifestyle trends of violence seen in younger men, could be related to higher A&E visits)
- Present findings through visually accessible means rather than technical modelling.

This topic also aligns strongly with group interest, as several members work in healthcare and understand the real-world implications of healthcare funding pressures and gender equity in care.

Target audience and technical level

Our project would primarily serve:

Primary audience

- NHS Managers, analysts and commissioners
- Healthcare professionals
- Decision-makers involved in funding and service planning

Secondary audience

- Public health researchers, policy teams
- Students and analysts exploring health datasets
- Members of the public interested in gendered health inequalities

Most of the audience will not be expected to have deep technical expertise; therefore, the report will focus on:

- Use of clear visualisations (bar charts, age-band comparisons, maps)
- Avoid heavy statistical jargon
- Focus on narrative insights such as:

“Women aged 20-39 experience higher healthcare costs largely due to maternity and reproductive care”

The dashboard will be visually led, accessible and explanatory rather than technical.

Datasets and anticipated challenges

At this early stage, we're open to acknowledging that our use of the datasets may change or vary as we progress further in making our arguments. So we've agreed for now that we will use

a combination of NHS England, ONS and potentially supplementary datasets to support our arguments (e.g. NHS Scotland) outlined below:

Primary Dataset:

“General health by age, sex and deprivation, England & Wales (Census 2021)”: Shows self-reported general health by sex and deprivation. Useful for SES (socio-economic status) + gender interplay.

<https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthandwellbeing/articles/generalhealthbyagesexanddeprivationenglandandwales/census2021>

“Hospital Admitted Patient Care Activity, 2023-24”: Data by age, sex, diagnosis, procedure. Great for gender differences in hospital care utilisation.

<https://digital.nhs.uk/data-and-information/publications/statistical/hospital-admitted-patient-care-activity/2023-24>

“Patient Level Activity and Costing, 2021-22”: Contains cost & activity data (acute, mental health, ambulance) – can compare male vs female cost usage.

<https://www.gov.uk/government/statistics/patient-level-activity-and-costing-2021-22>

Secondary Dataset: *These are supplementary to use as and when needed in aid of our arguments*

- Prescription costs <https://www.opendata.nhs.scot/dataset/prescriptions-in-the-community>
- A&E activity
<https://www.opendata.nhs.scot/dataset/weekly-accident-and-emergency-activity-and-waiting-times>
- GP payments
<https://www.opendata.nhs.scot/dataset/nhsscotland-payments-to-general-practice/resource/db3455ae-913e-4110-8439-d6a49164f7d9>
- Gender & age- A&E:
<https://digital.nhs.uk/data-and-information/publications/statistical/acute-patient-level-activity-and-costing/2018-19/age-and-gender>
- Healthcare expenditure, UK Health Accounts: 2021:
<https://www.ons.gov.uk/peoplepopulationandcommunity/healthandsocialcare/healthcaresystem/bulletins/ukhealthaccounts/2021>

Challenges and solutions

In no particular order

Issue 1: UK healthcare data is split by country

England has the richest datasets; Scotland and Wales use separate systems.

- ❖ **Solution:** Focus on NHS England and acknowledge limitations. Supplement with additional datasets, such as Scottish, if time allows.

Issue 2: Datasets may not cover identical years

The datasets may cover different time periods and not align to the same year

- ❖ **Solution:** Use the most recent datasets available and focus on trend insights rather than strict cross-year comparisons.

Issue 3: Costings in the dataset may not directly separate male vs female for all services

Some categories may be aggregated

- ❖ **Solution:** Use available age-band + sex combinations; where needed, estimate costs using activity count x average cost.

Issue 4: Non-binary and trans identities are not included

Most NHS cost datasets are sex - based only

- ❖ **Solution:** Acknowledge this limitation explicitly in the report

Issue 5: Outliers in healthcare cost data

Extreme cases may distort averages

- ❖ **Solution:** Inspect, document, and treat outliers appropriately (removal or separate analysis)

Team workflow, strengths, code management and timeline

How we plan to work and distribute tasks

We will distribute the workload based on each member's interests and strengths, guided by the poll we completed to identify preferred areas. Tasks will be assigned fairly and transparently, ensuring everyone contributes to the sections they feel most confident in.

To stay organised, we will use:

- **Trello** to track tasks, deadlines and progress
- **Weekly check-ins** to review updates and align priorities
- **Polly polls** for quick team decisions
- **Google Drive and Teams** for shared documents and communication

This structure ensures a balanced workload, consistent momentum and clear accountability

Team Strengths

- Strong industry insight from members working in healthcare
- A mix of creative, technical, and analytical thinking
- Diverse professional backgrounds, offering a wide range of expertise
- Strong organisation and communication, supported by established tools (Google Drive, Teams, Trello, Polly)
- A collaborative mindset and willingness to support one another

Team Weaknesses

- Varied levels of confidence with Python libraries (pandas, NumPy, matplotlib)
- Different personal schedules and workloads
- Coordination challenges due to evening classes and external commitments
- Not all members have healthcare or NHS Experience

We will address these weaknesses through shared learning, pairing on tasks, and maintaining flexibility in deadlines where possible.

How will we manage our code

- A **GitHub repository** has been created **to store** datasets, Jupyter Notebook files and supporting assets
- We will use **version control throughout**, with all changes committed to feature branches
- No code will be merged into the main branch without a **pull request and review** by the team (or majority)
- This ensures code quality, reproducibility and smooth collaboration

Expected Project Timeline

We will follow a structured workflow using Trello and simple milestone-style Gantt tracking:

Week 1-2:

- ☐ Finalise topic and hypothesis
- ☐ Draft and submit proposal
- ☐ Set up GitHub, a Trello board, and a shared workspace
- ☐ Collect datasets

Week 3:

- ☐ Begin data cleaning, joining, and preprocessing
- ☐ Identify trends, cost patterns, and key insights
- ☐ Conduct exploratory data analysis
- ☐ Build visualisations and a narrative dashboard

Week 4:

- ☐ Begin drafting the final report
- ☐ Final review, proofreading, and refinement
- ☐ Prepare presentation and submit final deliverables

This staged approach allows us to progress consistently, adapt to unforeseen issues and deliver a polished final product.

Why this project is valuable

- Addresses real-world concerns around NHS resource allocation and gender equity
- Provides actionable insights for policymakers and healthcare planners
- Combines health, data and social insight - highly engaging for the group
- *Allows us to apply Python, visualisation and analytical skills to a meaningful public issue*